

Operator MCQ Question

New symbols = [unit 2] = operator

Unary operator

(only one operand)

Increment

++

$x = 5$
 $x++$
 $x = x + 1$
 $= 5 + 1$
 $= 6$ (Math)

[value change by 1]

Decrement

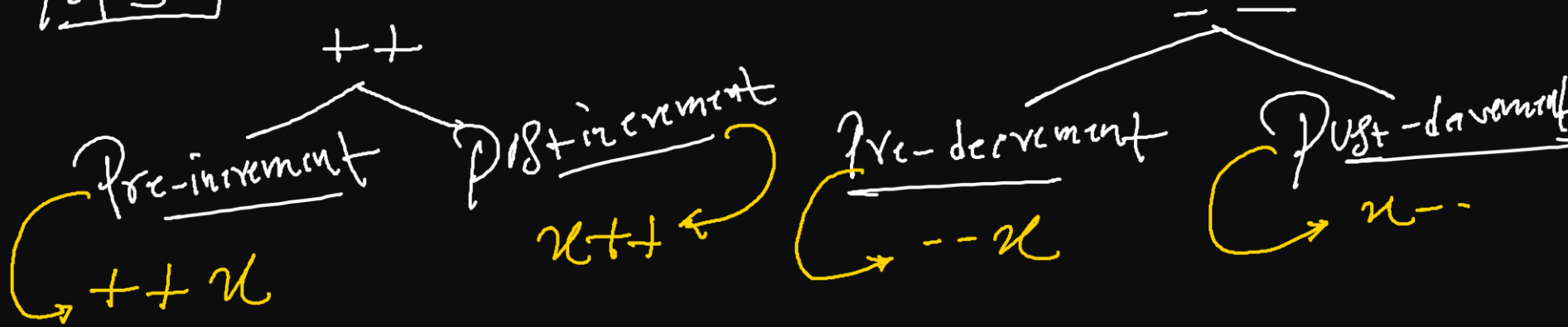
--

[value change by 1]

$x = 5$
 $x--$
 $x = x - 1$
 $= 5 - 1$
 $= 4$ (Math)

Operator MCQ Question

types



Operator MCQ Question

Pre => First change the original value and then use or assign the modified value on the variable memory.

int x = 5, y;

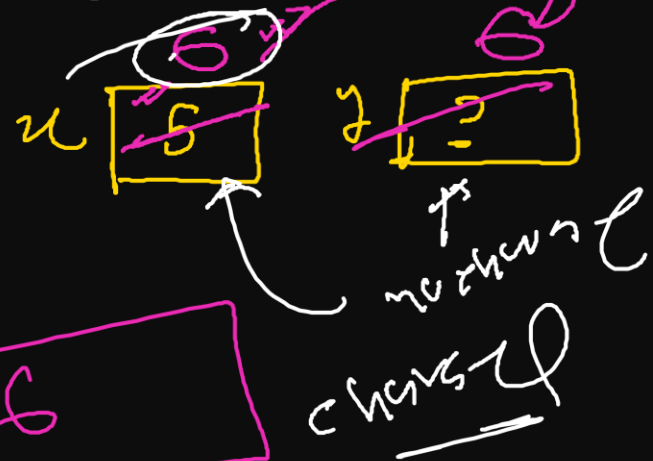
y = ++x;

printf("-x-2", y);

Pre-increment

printf("%d", y);

y = 6



Operator MCQ Question

++ = First original value return or used and then original value change

```
int x=10, y;
```

```
✓ y = x++;
```

```
pf(x)
```

```
pf(y)
```

++

x ~~11~~
10

y 10

x = 11
y = 10

ff ll
x=11
y=10
x=11

Operator MCQ Question

① $\text{int } n = 10$
 $x = n++$
 $\rightarrow \text{pre}(n) = 11$
 $+ n = 12$
 $\rightarrow \text{pre}(n) = 12$
 $n--$
 $--n = 10$
 $\rightarrow \text{pre}(n) = 10$
pre
ae

n ~~10~~ ~~11~~ ~~12~~ ~~11~~ 10

11 12 10

very important

Operator MCQ Question

①

```
int main()
{
    → int a = -3;
    → a = -a - a + !a;
    ← printf("%d\n", a);
    return 0;
}
```

6 ✓✓

a -3

$$\begin{aligned} a &= -a - a + !a \\ &= -(-3) - (-3) + !(-3) \\ &= 3 + 3 + 0 \\ &= 6 + 0 \\ &= 6 \end{aligned}$$

not $\rightarrow$ $!(-3)$
when any logical
any constant value then
that result
always 0 $\rightarrow$ either
+ve or -ve

$(!3) = 0$

Operator MCQ Question

2

```
int main()
{
    int a=2, b=1, c,d;
    c=a<b;
    d=(a>b) && (c<b);
    printf("c=%d, d=%d\n", c, d);
    return 0;
}
```

$a = 2$ $b = 1$ $c = 0$ $d = 1$

$c = a < b$
 $= 2 < 1$
 $c = \text{False}$ $c = 0$
 $c = 0$

$d = (a > b) \text{ and } (c < b)$
 $= 2 > 1 \text{ and } 0 < 1$
 $= 1 \text{ and } 1$
 $d = 1$

$c = 0$
 $d = 1$
ans

Operator MCQ Question

3

```
int main()
```

```
{
```

```
→ int a=5; ←
```

```
→ a=6;
```

```
a=a+5*a;
```

```
printf("a=%d\n",a);
```

```
return 0;
```

```
}
```

~~a = 5~~ 6 (initialization)
a = 6
assignment

$$\begin{aligned} a &= a + 5 * a \\ &= 6 + 5 * 6 \\ &= 6 + 30 \\ &= 36 \end{aligned}$$

ans

Operator MCQ Question

4

```
int main()
```

```
{
```

```
int a=5, b=5;
```

```
printf("%d, %d", ++a, b--);
```

```
printf("%d, %d", a, b);
```

```
printf("%d, %d", ++a, b++);
```

```
printf("%d, %d", a, b);
```

```
return 0;
```

```
}
```

logic change

Right to Left

← 6 5 →

①

b = 5

a = 5

b = 5

7, 5

Operator MCQ Question

```
5 int main() 5  
{  
  int x,y,z;  
  x=8++;  
  y=++x++;  
  z=(x+y)--;  
  printf("%d, %d, %d", x,y,z);  
  return 0;  
}
```

Compile time
never address
with constant value

Operator MCQ Question

```
int main() C
```

```
{
```

```
int a=4, b=8, c=3, d=9, z;
```

```
z = a++ + ++b * c-- --d;
```

```
printf("a=%d, b=%d, c=%d, d=%d, z=%d\n", a, b, c, d, z);
```

```
return 0;
```

```
}
```

$a = \cancel{4} 5$
 $b = \cancel{8} 9$ ✓
 $c = \cancel{3} 2$
 $d = \cancel{9} 8$

$z = 23$

$a = 5$

$b = 9$

$c = 2$

$d = 8$

$z = 23$

$$z = a++ + ++b * c-- --d$$
$$= 4 + 9 * 3 - 8$$

$$z = 4 + 27 - 8$$
$$= 31 - 8$$

23

int main()

{

int a = 14, b, c;

a = a % 5;

b = a / 3;

c = a / 5 % 3;

printf("a = %d, b = %d, c = %d\n", a, b, c);

return 0;

}

ans

Will 0

a = 14 if b = 1 c = 0

a = a % 5 = 14 % 5

5 | 14 | 2 = 4
10
4 ← Remainder

b = a / 3 = 14 / 3 = 4
3 | 4 | 1
3
1

5 | 4 | 0
5
0

c = a / 5 % 3
= 4 / 5 % 3
= 0 % 3
c = 0

Operator MCQ Question

```
int main()
```

```
{
```

```
    int x,y,z,k=10;
```

```
    k += (x=5,y=x+2, z=x+y);
```

```
    printf("x=%d, y=%d, z=%d, k=%d\n",x,y,z,k);
```

```
    return 0;
```

```
}
```

$u = 5$

$y = 7$

$z = 12$

$k = 10$

5, 7, 12, 22

22

$k += \text{_____}$

$$\begin{aligned} \Rightarrow k &= k + (u=5, y=u+2, z=u+y) \\ &= 10 + (\cancel{u=5}, \cancel{y=7}, \textcircled{z=12}) \\ &= 10 + 12 \\ &= 22 \end{aligned}$$

Right to Left

Operator MCQ Question

int main()

(a)

{

int a = 15, b = 13, c = 16, x, y;

✓ $x = a - 3 \% 2 + c * 2 / 4 \% 2 + b / 4;$

~~$x = a - 3 \% 2 + b / 4;$~~

printf("x=%d, y=%d\n", x, y);

return 0;

}

a = 15

b = 13

c = 16

x =

y =

$$x = 15 - \underbrace{3 \% 2}_{(1)} + \underbrace{16 * 2 / 4 \% 2}_{\substack{2 \times 4 = 8 \\ 8 / 4 = 2 \\ 2 \% 2 = 0}} + \underbrace{b / 4}_{13 / 4 = 3}$$

$$= 15 - 1 + 0 + 3$$

$$= 14 + 3 = 17 \text{ ans}$$

Operator MCQ Question

compiler check

int a = 10

printf(a++, --a, ++a, a)

① undefined behaviour

why

diff?

There is no standard rule on a programmer to create the inside the operator the printf the

VS code = ?
turbo = ?

Operator MCQ Question

a = 10 ~~++~~ 11 posts
P ↓ C at + , -- a , ++ a , 9
VS code
~~10 11 12 11 10~~
[10] a++
--a ++a
[10, 10, 11, 10] [turbo]

